

Big Byte

The computing highway is a 64-bit lane and your machine is powered by ClawHammer. Itanium takes the backseat. Ready?

For three decades (and counting), the speedy growth of the microprocessor has led to some electrifying breakthroughs in computing. With each evolutionary step forward, a single telling yardstick—power—gauges the progress of these electronic brains. With yet another evolutionary leap in the offing, PCs are set to undergo a change like no other. Computing power will speak of new dimensions to explore and exploit.

Undoubtedly, 64-bit has the power to steer us towards futuristic ideas, bringing them closer than we ever thought possible. For instance did you ever imagine 100 percent accurate voice recognition? But with recent developments, computers will not only hear you, they'll understand your every word, no matter how unique your speech. And it's not only the complex routines such as these that need boosts of processing power. The requirement starts from the lowest factor in the computing chain: the common user.

Without the power of 32-bit computing, the ease with which we use a PC

via operating systems such as Windows, could never have been possible. The next chapter in computing comes in the form of 64-bit computing. It's not only a lot more powerful than the existing 32-bit technology, but it also opens doors for turning today's obstacles into tomorrow's realities.

So what is 64-bit computing? It's certainly not something you can touch or see—it's something you can experience. It all begins beneath the hood, the heart of all computers, the CPU. Today's processors operate at 32-bit, giving them room to digest 32-bits of information at once. The oncoming 64-bit onslaught will obviously have twice the capacity. Think of it as a 32-lane highway doubling its width; it's a tremendous change in computing power!

But in order to cash in on the promises of a brilliant future, we need the keys to unlock it. The first and most fundamental obstacle rearing itself in our paths is, what else, software support. There are other gaps too that make the transition from 32-bit to 64-bit not exactly a cakewalk. Essentially, 64-bit computing is a

delicate matter wherein the entire balance of the computing industry is at stake. At least this is no exaggeration!

Why all the hoopla?

The two biggest players in the CPU market, Intel and AMD, are battling it out with each other to reach one common goal. Who'll control the developing 64-bit computing market in the coming years?

Each of these companies desire to set a paradigm for the industry to follow. The score so far: Intel quietly made it out first with their Itanium chip, a high-end processor dedicated to server and enterprise level computing. Built from the ground up, Itanium is completely focussed on the 64-bit computing platform, a strategy that could possibly be the bane of its entire existence. By jumping the gun directly to a platform that has little support, Intel has shut out more than 93 percent of existing applications, resources and technology. This leaves users—like us in India—in a lurch. Sure, we can purchase a top-of-the-line chip, but without the right enabling tools, what could we possibly do with an Itanium? AMD, conscious of this,

